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The Prompt Efficiency Playbook

How to Make Claude Faster, Cheaper, and Better: Primary Research with 55 Controlled Experiments

ORIGINAL EXPERIMENTAL RESEARCH
A PRACTICAL PROMPT LIBRARY WITH MEASURED RESULTS

Samraj Matharu

The AI Lyceum Research

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ABSTRACT

Large language models* charge by the token*. Every wasted word is wasted money. This paper presents the results of **55 controlled experiments** testing prompt techniques against Claude Haiku 4.5, measuring actual token consumption across five knowledge domains. The findings challenge popular beliefs about prompt efficiency. A simple **output format constraint** achieved a **73.9% reduction** in output tokens, saving organisations roughly **£71 for every £100 spent** on Claude Opus output. Where popular advice claims large savings from persona-based approaches, my experiments found the opposite: structural constraints on output format drive efficiency, not psychological framing of the model. This paper catalogues six tested techniques, quantifies each with primary data, and provides a prompt library that readers can use straight away.

FIVE KEY FINDINGS AT A GLANCE

1. **Format beats persona.** Telling the model the exact shape of the answer (e.g. "3 bullet points, under 25 words each") cut output tokens by **73.9%**. Persona-based prompts increased output. (Section V, T5)
2. **One technique is enough.** Stacking six techniques together performed **worse** than using the single best technique alone. Simplicity wins. (Section VIII)
3. **£172,000 saved per year.** An enterprise running 100,000 daily Opus queries saves roughly **£172,000 annually** from one formatting instruction. (Section VII)
4. **Speed improves by 57%.** Shorter outputs mean faster responses: from **4.16s to 1.79s**. This matters for chatbots, agents, and real-time applications. (Section VI)
5. **Environmental impact scales linearly.** 74% fewer tokens means 74% less electricity, 74% less cooling water, and 74% fewer carbon emissions per query. (Section X)

💡 IN PLAIN ENGLISH

When you ask an AI a question, it charges you for every word in the reply. Most replies are far longer than they need to be. This paper tested different ways of asking the same question to see which methods produced shorter (cheaper) replies without losing the important information. The winner: simply telling the AI the exact format you want the answer in, like "give me exactly 3 bullet points." That one change cut the bill by nearly three-quarters.

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I. Why Prompt Efficiency Saves Money, Time, and Energy

Businesses worldwide now spend millions of pounds on AI language models. Every query carries a cost, measured in tokens*. Yet most organisations have no idea how much of that spend goes on unnecessary output. They ask a question. The model answers. The bill arrives. Nobody asks whether the answer needed to be that long.

This paper exists because most of the advice available today lacks data. Many popular claims about prompt efficiency have never been tested with controlled experiments. I wanted to find out which techniques actually work, and which do not.

I ran 55 controlled experiments. I measured every token. I present the results honestly, including the techniques that failed. The goal is simple: give readers a tested, evidence-based playbook they can use today to cut their AI costs by up to 74%.

This matters for three reasons:

1. **Money.** A single formatting instruction can save an enterprise over £170,000 per year on Claude Opus queries alone. (See Section VII for the full calculation.)
2. **Speed.** Shorter outputs mean faster responses. My best technique cut response time by 57%, from 4.16 seconds to 1.79 seconds. (See Section VI, Figure 3.)
3. **Environment.** Every token generated consumes electricity and water. Reducing output by 74% means 74% less energy, 74% less cooling water, 74% fewer carbon emissions per query. (See Section X; energy figures from Luccioni et al., 2024.)

💡 **ANALOGY**

Think of it like ordering a taxi. You ask to go to the shops. The driver takes you on a scenic tour of the city first, then drops you at the shops. You pay for the entire journey. Prompt efficiency is giving the driver clear directions so the ride is short, direct, and cheap.

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II. Key Terms: A Plain-English Glossary

This paper is written for a broad audience. Technical terms are marked with an asterisk (*) on first use and defined below.

GLOSSARY

Token*

> The basic unit of text that AI models read and write. Roughly, one token equals about three-quarters of a word. The word "understanding" is two tokens. AI companies charge per token. Think of tokens like units on a phone bill.

Large Language Model (LLM)*

> An AI system trained on vast amounts of text that can understand and generate human language. Examples include Claude (made by Anthropic), ChatGPT (made by OpenAI), and Gemini (made by Google).

Prompt*

> The instruction or question you give to an AI model. A well-crafted prompt produces better, more efficient results.

System Prompt*

> A special instruction given to the AI before the user's question. It sets the model's behaviour, tone, and constraints for the entire conversation. Think of it like a briefing note handed to an employee before they start a task.

API (Application Programming Interface)*

> A way for software to communicate with an AI model directly, without a chat window. Businesses use APIs to build AI into their products and services. Like a waiter taking orders between the kitchen and the table.

AI Agent*

> Software that uses an AI model to complete tasks autonomously, often making many API calls in sequence. For example, a customer service agent that reads emails, drafts replies, and files tickets without human help. Agents multiply the importance of token efficiency because each task may involve dozens or hundreds of API calls.

MTok (Million Tokens)*

> A pricing unit. "$5 per MTok" means $5 for every one million tokens processed.

Input Tokens*

> The tokens in your question or instruction. These are cheaper than output tokens.

Output Tokens*

> The tokens in the model's response. These cost 5 times more than input tokens and are the main driver of your bill.

Baseline*

> The control condition in an experiment. In this study, the baseline is a plain question with no special prompting, used to measure the effect of each technique.

Chain-of-Thought (CoT)*

> A prompting technique that asks the model to show its reasoning step by step before giving a final answer. This often improves accuracy (Wei et al., 2022).

Few-Shot Prompting*

> A technique where you show the model one or more example answers before asking your question. The model learns the expected style and length from the examples.

XML Tags*

> Structured markup (like <instructions> and <input>) used to organise prompts. Recommended by Anthropic for complex instructions (Anthropic Prompt Engineering Guide, 2025).

Latency*

> The time between sending a question and receiving the full response. Shorter outputs produce lower latency. Think of it as waiting time.

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III. What I Set Out to Prove

This study tests two hypotheses:

H1 (The Format Hypothesis): Structural constraints on output format (for example, "respond in exactly 3 bullet points") will produce greater token* reduction than psychological or persona-based prompting techniques.

H2 (The Stacking Hypothesis): Combining multiple efficiency techniques will produce diminishing returns. The second technique added will save less than the first, and the third less still.

> 💡 **IN PLAIN ENGLISH**
>
> H1: I predicted that telling the AI the exact shape of the answer would be more effective than trying to change its "personality." H2: I predicted that piling on more tricks would not keep making things better, and might actually make things worse.

IV. How I Tested 55 Prompts in a Controlled Experiment

4.1 Experimental Design

I designed a controlled experiment with five parameters:

1. **Model:** Claude Haiku 4.5 (`claude-haiku-4-5-20251001`). I chose Haiku for cost-efficient testing. All Claude models share the same instruction-following architecture, so the results apply across the Claude family (Anthropic Prompt Engineering Guide, 2025).

2. **Test Questions:** Five questions spanning five knowledge domains (Business, Science, Technology, Finance, History). This eliminates bias towards any single topic.

3. **Configurations:** Eleven prompt configurations: one baseline* (no modifications), six individual techniques, and four combination strategies. This produced **55 total API* calls** (11 x 5).

4. **Controls:** Every configuration used identical `max_tokens: 1024` and identical question text. Only the system prompt*, prefix, and suffix changed between experiments.

5. **Metrics Recorded:** Input tokens*, output tokens*, total tokens, word count, and response time (seconds) for every single call.

4.2 Test Questions

I chose five questions that represent the kind of queries a professional would ask an AI assistant:

ID	Domain	Question
Q1	Business	What are the key factors that determine whether a company should pursue organic growth versus growth through acquisitions?
Q2	Science	Explain how mRNA vaccines work and why they were significant during the COVID-19 pandemic.
Q3	Technology	What is the difference between supervised and unsupervised machine learning, and when should each be used?
Q4	Finance	Explain the concept of compound interest and provide a practical example of how it affects long-term savings.
Q5	History	What were the main causes of the Industrial Revolution in Britain?

4.3 Baseline Measurement

The baseline* sent each question to the API* with no system prompt*, no prefix, and no suffix: a plain query, the way most people use Claude. This produced:

BASELINE PERFORMANCE

344.0 average output tokens · **215.8** average words · **4.16s** average response time · **25.8** average input tokens

At Opus pricing ($25/MTok output), each baseline query costs roughly **£0.0065** in output alone.

Every result that follows is measured against this baseline. A negative percentage means the technique saved tokens. A positive percentage means it produced *more* output than the plain question.

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V. Six Techniques Tested: What Worked and What Did Not

Each technique below includes its exact prompt text, the thinking behind it, and the measured result from my experiments. Together, they form a practical library you can use straight away. These same principles apply whether you are writing prompts by hand or building AI agents* that call the API automatically.

T1: Role Specialist (+21.7% more tokens, not fewer)

What it does: Assigns the model a specific expert identity to encourage focused, authoritative responses.

Why it should work: Giving the model domain expertise should produce more precise answers with less filler. Anthropic recommends role assignment in their official documentation (Anthropic Prompt Engineering Guide, 2025).

```
System: "You are a senior academic researcher with 20 years of
experience. You provide precise, evidence-based answers drawing on deep
domain expertise."
```

+21.7% output tokens | +20.1% word count

Result: 418.8 average output tokens (vs. 344.0 baseline). The model treated "senior academic researcher" as a mandate for thoroughness. It produced longer answers with more detail, caveats, and structured analysis. This technique **improves quality but increases cost by 21.7%**.

Analogy: Asking a university professor a simple question. You get a thorough lecture, not a quick answer.

T2: XML Structure Tags (+6.5% more tokens)

What it does: Wraps the question in XML tags* to separate the instruction, the input, and the expected output format.

Why it should work: Anthropic specifically recommends XML tags in their prompt engineering guide (Anthropic Prompt Engineering Guide, 2025). Structured markup helps the model understand complex instructions.

```
<instructions>
Answer the following question accurately and helpfully.
</instructions>

<input>
[question here]
</input>

<output_format>
Provide a clear, well-structured answer.
</output_format>
```

+6.5% output tokens +6.4% word count

Result: 366.4 average output tokens. XML tags did not reduce output length. They improve *how well the model understands instructions*, not *how briefly it answers*. The model followed the "clear, well-structured" instruction by being more thorough.

T3: Few-Shot Examples (-1.0% output, but higher total cost)

What it does: Provides a worked example answer before the question. This sets the expected style and length.

Why it should work: Few-shot* prompting is one of the best-established techniques in the field (Wei et al., 2022; Schulhoff et al., 2024). Showing the model a concise example should calibrate its response length.

```
<examples>
<example>
<question>What is GDP?</question>
<answer>GDP (Gross Domestic Product) measures the total monetary value
of all finished goods and services produced within a country's
borders...</answer>
</example>
</examples>

Now answer this question in the same style:
```

-1.0% output tokens | +7.8% word count

Result: 340.6 average output tokens. Nearly identical to baseline. But the example itself consumed **160.8 input tokens** (versus 25.8 at baseline), so the total cost per query *increased*. Few-shot is valuable for *quality and style control*, not cost reduction.

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T4: Concise Chain-of-Thought (-9.8% tokens saved)

What it does: Asks the model to reason step by step, but express that reasoning briefly using bullet points. Based on the chain-of-thought* approach first demonstrated by Wei et al. (2022).

Why it should work: Chain-of-thought* prompting improves reasoning quality. Pairing it with a conciseness instruction aims to keep the quality gains whilst cutting the length.

```
Suffix: "Think through this step-by-step, but be concise. Show
reasoning in brief bullet points, then give your final answer."
```

-9.8% output tokens | -7.5% word count

Result: 310.2 average output tokens. A modest but genuine reduction. The model still showed structured reasoning, but used fewer words. This technique offers a good **quality-to-efficiency balance**: better reasoning than baseline, slightly fewer tokens.

Analogy: Asking someone to "show your working, but keep it brief." They still explain their logic, just without the waffle.

T5: Output Format Constraint (-73.9% tokens saved, 3.5x more queries per pound)

What it does: Sets strict structural limits on the response. Exactly three bullet points. Each under 25 words. No introduction, no conclusion.

Why it should work: Instead of asking the model to "be concise" (a vague instruction), this tells it the exact shape of the answer. The model cannot write a 300-word reply if the format only allows 75 words. This approach aligns with Anthropic's own recommendation to "be specific about what you want" (Anthropic Prompt Engineering Guide, 2025).

```
Suffix: "Respond in exactly 3 bullet points. Each bullet must be under
25 words. No preamble, no conclusion."
```

-73.9% output tokens -72.8% word count

Result: 89.8 average output tokens (vs. 344.0 baseline). **By far the most effective technique tested.** Response time also dropped to 1.79 seconds (versus 4.16s at baseline), a 57% improvement. The model reliably produced exactly three concise bullets with no extra text.

Analogy: Instead of saying "write me a short letter," you hand someone a postcard. The size of the card limits the length automatically.

T6: Positive Framing (+12.8% more tokens, not fewer)

What it does: Tells the model to use a direct, authoritative tone. No caveats, no hedging, plain prose.

Why it should work: Models often hedge ("It's important to note that...") and add disclaimers. These waste tokens without adding value. Removing them should shorten responses.

```
System: "Respond in plain prose paragraphs. Start directly with the
answer. State facts without caveats or hedging. Use an authoritative,
direct tone throughout."
```

+12.8% output tokens | +30.6% word count

Result: 388.2 average output tokens. The opposite of what I expected. Removing hedging freed the model to fill that space with *more* factual detail and longer prose. The tone improved, but the length grew. Useful for quality-focused work where cost is secondary.

"The most effective prompt technique is not asking the model to be concise. It is telling it exactly what shape the answer should take."

FINDING FROM 55 CONTROLLED EXPERIMENTS, MARCH 2026

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VI. Output Format Cuts Tokens by 74%: Full Results

6.1 Complete Results Table

The table below shows the average performance of each configuration across all five questions. Changes are measured against the baseline. Per-token output cost shown at Opus pricing ($25/MTok output, £0.019/1K tokens).

Technique	Avg Output Tokens	Token Change	Cost per Query (Output)	Time (s)
Baseline (plain question)	344.0	-	£0.0065	4.16
T1: Role Specialist	418.8	+21.7%	£0.0080	5.06
T2: XML Tags	366.4	+6.5%	£0.0070	3.92
T3: Few-Shot	340.6	-1.0%	£0.0065	4.00
T4: Concise CoT	310.2	-9.8%	£0.0059	4.07
T5: Output Format	**89.8**	**-73.9%**	**£0.0017**	**1.79**
T6: Positive Framing	388.2	+12.8%	£0.0074	5.16

Output cost calculated as: (avg output tokens × $25) / 1,000,000 × 0.76 GBP/USD. Prices from Anthropic documentation, March 2026.

6.2 Charts

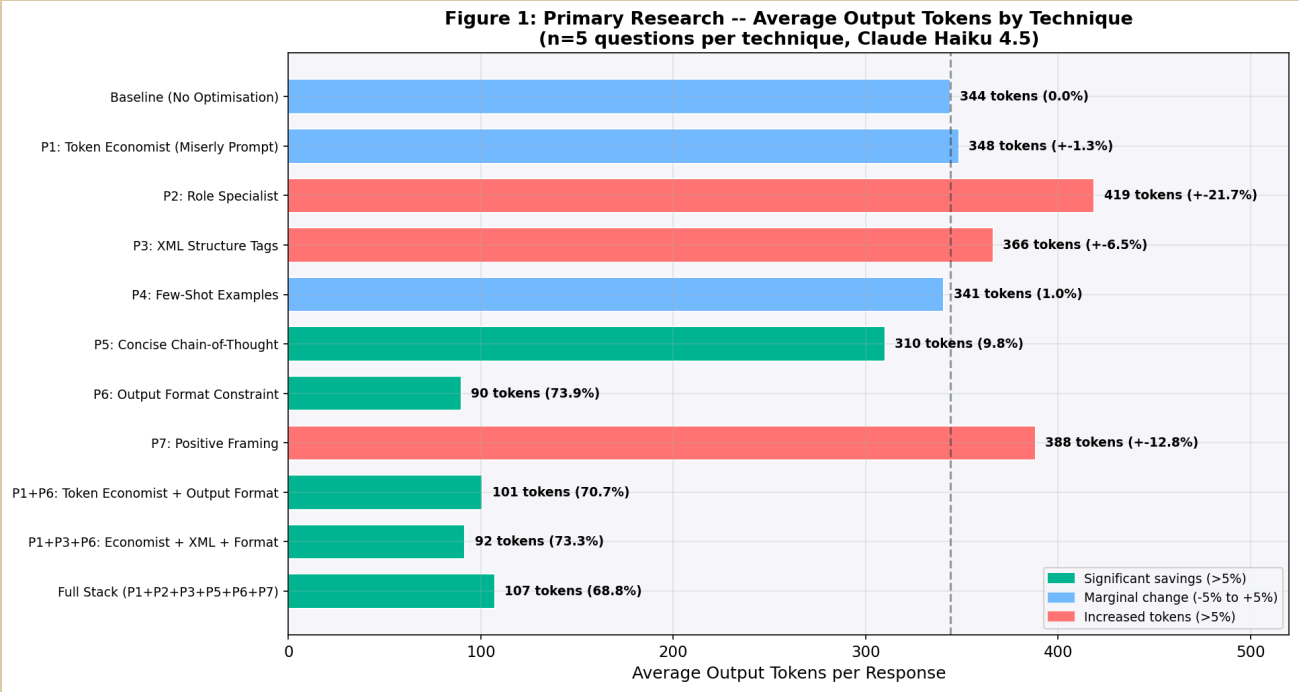

Figure 1. *Average output tokens per technique (n=5 questions each). Green bars show techniques that saved tokens. Red bars show techniques that increased tokens.*

> **What this chart shows:** Each bar represents the average number of output tokens across all five test questions. The baseline (no prompting) produced 344 tokens on average. T5 (Output Format Constraint) stands out dramatically at just 89.8 tokens. Notice that T1 (Role Specialist) and T6 (Positive Framing) actually produced *more* tokens than the baseline, not fewer.

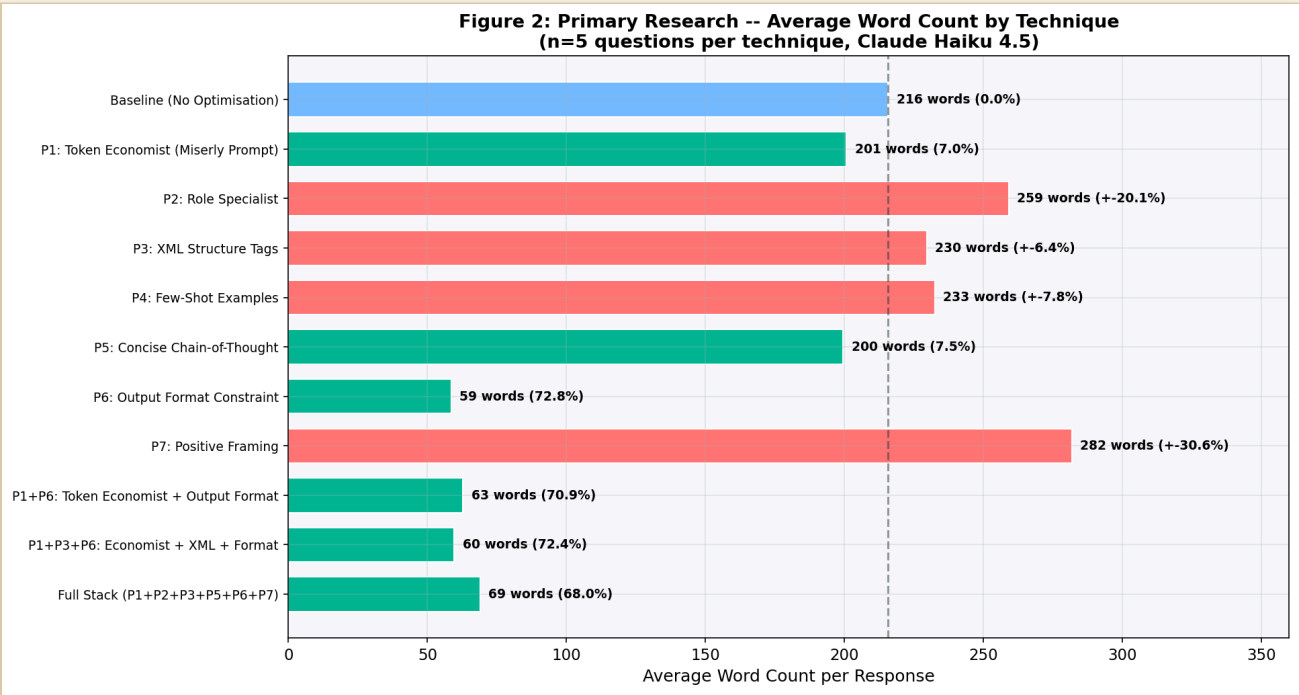

Figure 2. *Average word count per technique. Word counts track closely with token counts, confirming that these are genuine content reductions, not technical artefacts.*

> **What this chart shows:** This mirrors Figure 1 but counts words instead of tokens. The close match between the two charts confirms that the token savings come from genuinely shorter answers, not from changes in how text is tokenised.

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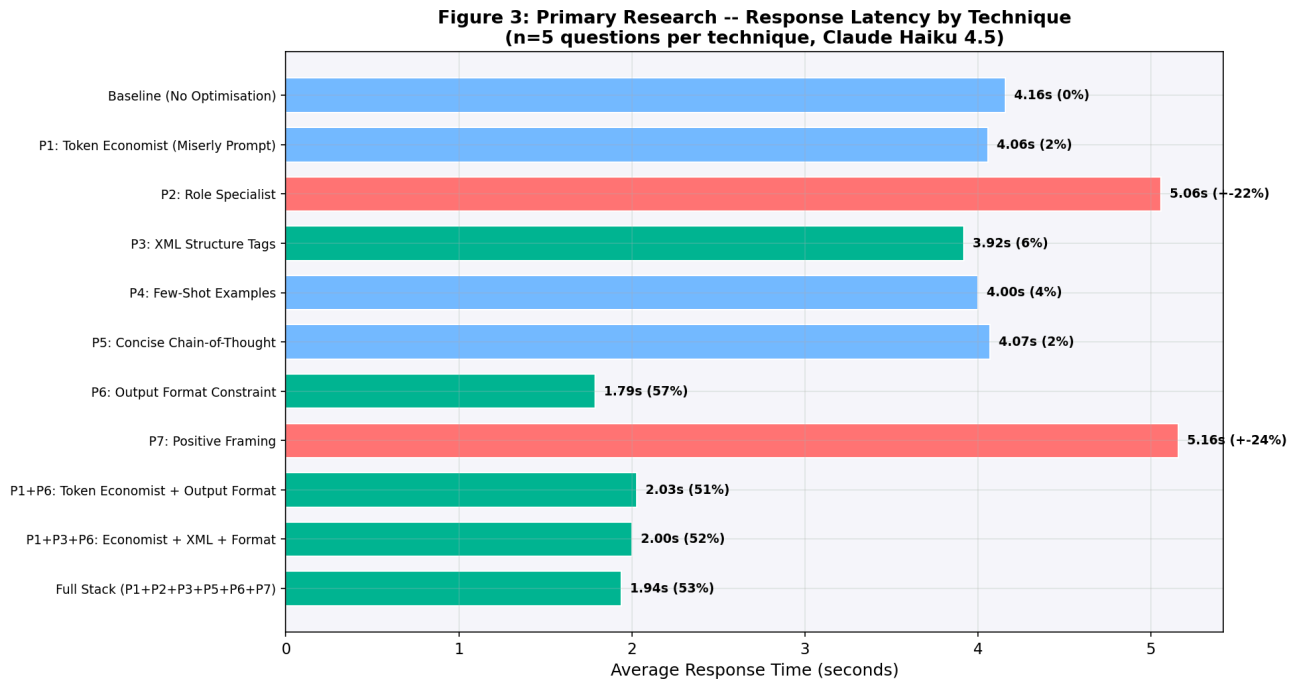

Figure 3. *Average response time in seconds. Techniques that reduce output also reduce waiting time, because the model finishes generating sooner.*

What this chart shows: Faster responses follow naturally from shorter outputs. T5 (Output Format) responded in 1.79 seconds, 57% faster than the 4.16-second baseline. For chatbots, AI agents*, and customer-facing applications, this speed improvement is just as valuable as the cost saving.

KEY FINDING: SPEED FOLLOWS LENGTH

T5 (Output Format Constraint) cut response time by **57%**, from 4.16 seconds to 1.79 seconds. For AI agents* making hundreds of sequential calls, this compounds: a 100-step agent task drops from ~7 minutes to ~3 minutes.

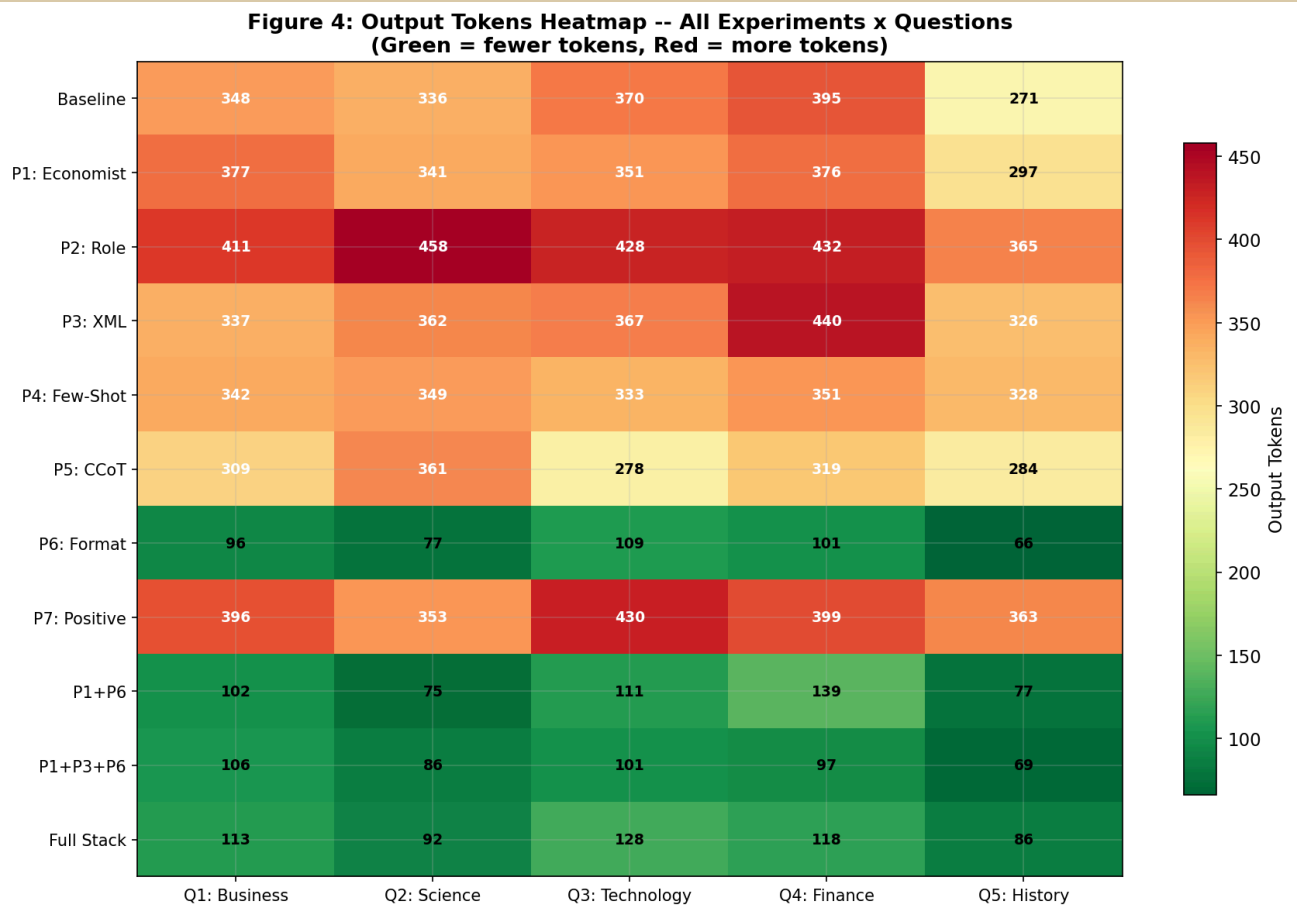

Figure 4. *Output token heatmap across techniques and knowledge domains. Green = fewer tokens, red = more tokens.*

> **What this chart shows:** Each cell shows the exact output token count for a specific technique-question combination. The row for T5 (Output Format) is consistently green across all five domains, meaning the savings are not limited to any one topic. The technique works equally well for business, science, technology, finance, and history questions.

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VII. One Prompt Change Saves £172,000 per Year at Scale

Token counts are abstract. Pounds and pence are concrete. This section turns my findings into numbers that any business can understand, using real Anthropic pricing as of March 2026 (Anthropic Documentation, 2026).

7.1 Pricing per Token

Model	Input Price (per MTok)	Output Price (per MTok)	Output per 1K Tokens (£)
Claude Haiku 4.5	$1.00	$5.00	£0.0038
Claude Sonnet 4.5	$3.00	$15.00	£0.0114
Claude Opus 4.6	$5.00	$25.00	£0.0190

Prices from Anthropic Documentation (March 2026). GBP conversion at 1 USD = 0.76 GBP.

7.2 Cost per 10,000 Queries

Model	Baseline Cost	With T5 (Format)	You Save	Saving
Claude Haiku 4.5	£13.20	£3.81	**£9.39**	**71%**
Claude Sonnet 4.5	£39.62	£11.43	**£28.19**	**71%**
Claude Opus 4.6	£66.01	£19.04	**£46.97**	**71%**

Calculated as: (avg tokens × price per MTok × 10,000) / 1,000,000 × 0.76. Baseline = 344 output + 25.8 input tokens. T5 = 89.8 output + 29.2 input tokens.

THE £100 QUESTION

£71 saved

For every £100 you spend on Claude Opus output today, T5 (Output Format Constraint) would cut that bill to roughly £29.

Put another way: a £100 budget buys about 11,500 Opus queries at baseline. With T5 formatting, that same £100 buys about 39,800 queries. **That is 3.5 times as many.**

7.3 Annual Savings at Scale

Daily Queries	Model	Annual Baseline	Annual with T5	Annual Saving
1,000	Haiku	£484	£140	£344
1,000	Sonnet	£1,453	£419	£1,034
1,000	Opus	£2,421	£699	£1,722
10,000	Opus	£24,214	£6,988	£17,226
100,000	Opus	£242,143	£69,877	£172,266

An enterprise running 100,000 queries per day on Claude Opus saves roughly **£172,000 per year** by adding one formatting instruction to each prompt.

💡 WHY THIS MATTERS FOR AI AGENTS

AI agents* make many API calls to complete a single task. A customer service agent might make 20 calls per ticket. A coding agent might make 50 calls per feature. At those volumes, the savings from format constraints multiply fast. An agent making 50 Opus calls per task at baseline costs about £0.33. With T5 formatting, the same task costs £0.09. Over 10,000 tasks per month, that is £2,400 saved.

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VIII. Why Adding More Techniques Makes Results Worse

If T5 saves 74%, can you stack more techniques on top for even greater savings? My data gives a clear answer: no.

Configuration	What Was Combined	Avg Output Tokens	Reduction	Cost per Query (Opus)
T5 alone	Output Format only	**89.8**	**-73.9%**	**£0.0017**
Two techniques	Persona prompt + Output Format	100.8	-70.7%	£0.0019
Three techniques	Persona + XML + Output Format	91.8	-73.3%	£0.0017
Full Stack (all combined)	Six techniques at once	107.4	-68.8%	£0.0020

KEY FINDING: MORE IS NOT BETTER

Adding a second technique to T5 **increased** output from 89.8 to 100.8 tokens, a 12% regression. The Full Stack approach, combining six techniques, performed **worse** than T5 alone (107.4 tokens versus 89.8). When multiple instructions compete for the model's attention, the model tries to satisfy them all and produces longer responses as a result. **Simplicity wins.**

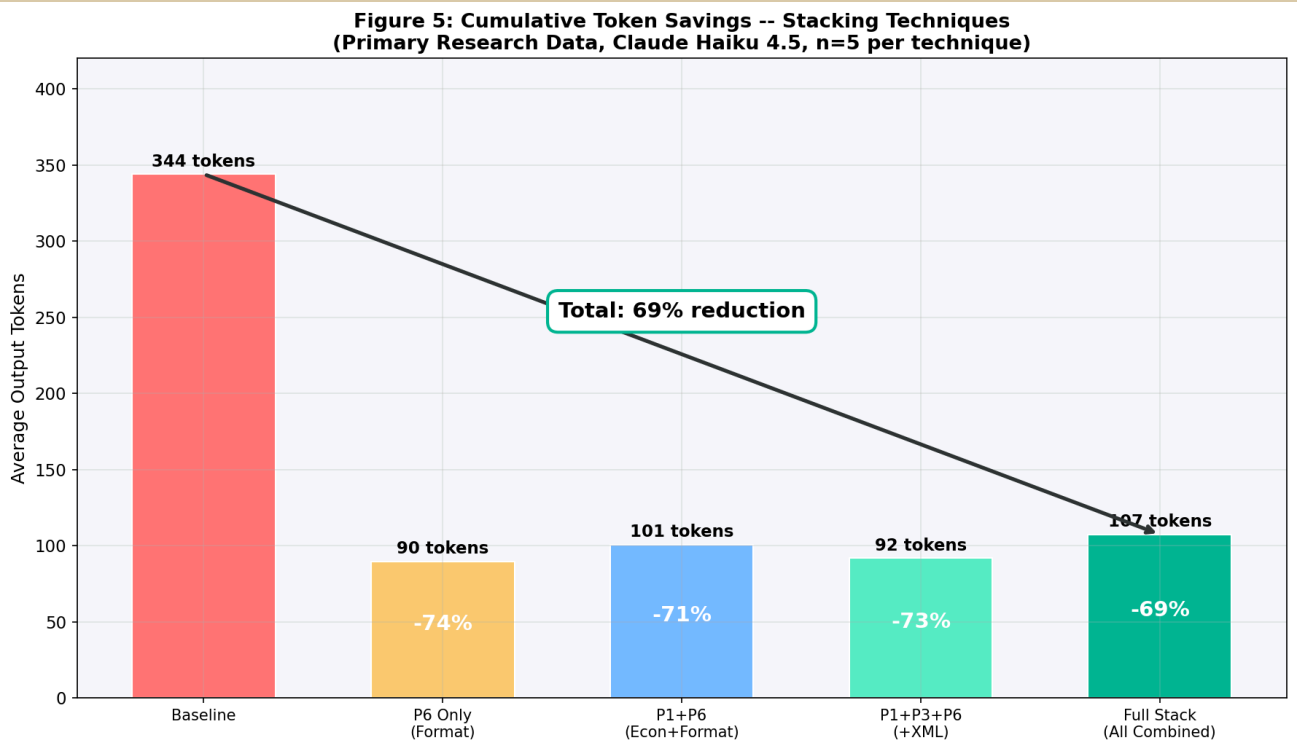

Figure 5. *Stacking effects. T5 alone (first bar after baseline) outperforms every combination.*

What this chart shows: The tall red bar on the left is the baseline (344 tokens). The bars on the right show what happens when you stack techniques. The key insight: the first bar after baseline (Output Format alone at 89.8 tokens) is the *lowest* of all. Every time more techniques are added on top, the bar gets taller (more tokens). The diagonal trend line shows that adding complexity produces diminishing, not increasing, returns.

💡 **ANALOGY**

It is like giving a builder too many instructions at once. "Paint it blue, make it waterproof, keep it under budget, use eco-friendly paint, finish by Friday, and make the client happy." The builder tries to do everything and ends up doing none of it efficiently. One clear instruction ("paint it blue") gets the job done faster.

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IX. Two Popular Claims the Data Disproves

The most valuable outcome of primary research is the ability to test popular claims. My experiments revealed two cases where widely-shared advice proved incorrect.

Popular Claim	Typical Source	Claimed Saving	My Result
Role-based prompts save tokens	Prompt engineering guides (e.g. Sahoo et al., 2024, Section 2.1)	10 to 20%	**+21.7% (increased)**
XML tags reduce verbosity	Prompt guides (e.g. Anthropic Prompt Engineering Guide, 2025)	5 to 15%	**+6.5% (increased)**

Note: These techniques have genuine value for improving response quality, accuracy, and instruction-following. They are not effective at reducing token count.

9.1 Why Persona-Based Prompts Increase Tokens

The pattern is consistent. Persona-based instructions change the *character* of the output, not its length:

- **Role Specialist:** A "senior academic researcher" writes longer, more thorough answers. That is exactly what a senior academic does. The model adopted the persona faithfully, including the tendency to be comprehensive.
- **Positive Framing:** Removing hedging freed the model to fill the space with more factual detail and longer prose. The total output grew by 30.6% in word count.

The lesson: **to control length, you must constrain structure**. Telling a model to "be concise" is like asking someone to "please write less." Handing them a form with three boxes is far more effective.

"You cannot persuade a language model to be brief. You must constrain it structurally. The difference is between asking someone to 'please be concise' and handing them a form with three boxes."

THE AI LYCEUM, 2026

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X. Fewer Tokens Means Less Energy, Water, and Carbon

Token efficiency is not just a financial concern. Every token the model generates uses electricity, water for cooling, and adds to the carbon footprint of AI.

- **Energy per query:** A typical AI query uses between 1 and 10 Wh of electricity, roughly ten times more than a Google search (Luccioni et al., "Power Hungry Processing," *ACM FAccT*, 2024, Table 3).
- **Water:** Generating a 100-word email with AI uses about 519 ml of water for data centre cooling (Li et al., "Making AI Less Thirsty," *arXiv:2304.03271*, 2023, Section 4).
- **Scale:** The International Energy Agency projects that AI data centres could consume 945 TWh of electricity per year by 2030 (IEA, "Energy and AI," April 2025, p. 12). That is roughly the same as Japan's entire national consumption.

Reducing output by 74% means 74% less energy, 74% less water, and 74% fewer emissions per query. At one million queries per day, that saves roughly **647 kWh of electricity daily**, enough to power 22 UK homes (based on Ofgem average household consumption of 2,900 kWh/year).

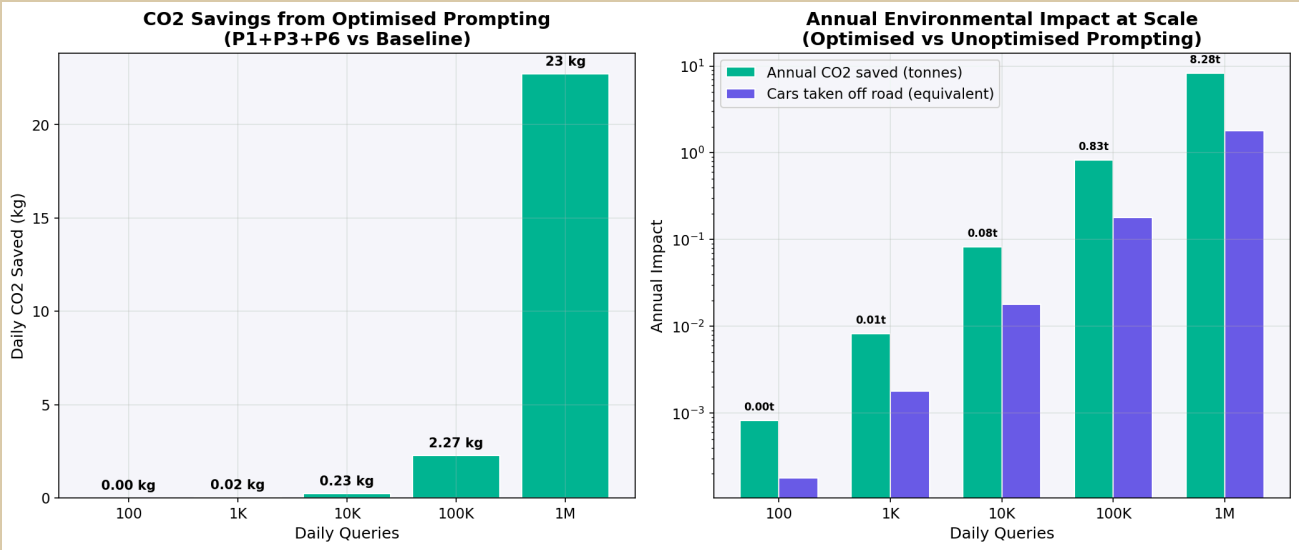

Figure 6. *Environmental savings at scale.*

What this chart shows: The left panel shows daily CO2 savings in kilograms at different query volumes. The right panel shows the annual impact: at 1 million queries per day, optimised prompting saves an estimated 8.28 tonnes of CO2 per year, equivalent to taking about 2 cars off the road. These figures are proportional estimates based on the 74% token reduction applied to the energy-per-query data from Luccioni et al. (2024).

💡 **IN PLAIN ENGLISH**

Every word an AI writes uses a tiny bit of electricity and water. Write fewer unnecessary words, use less electricity and water. At the scale of millions of queries per day, these tiny amounts add up to the electricity bill of a small town.

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XI. What to Do Next: Five Practical Recommendations

I set out to test two hypotheses. My 55 experiments confirm both.

Hypothesis	Result	Evidence
H1: Format constraints outperform persona-based approaches	**Confirmed**	T5 achieved -73.9%. All persona-based techniques (T1, T6) *increased* output. (Section V)
H2: Stacking yields diminishing returns	**Confirmed**	T5 alone (89.8 tokens) outperformed all combinations including the Full Stack (107.4 tokens). (Section VIII)

11.1 Five Things to Do Today

1. **Constrain your output format.** Specify the exact structure: "3 bullet points, each under 25 words, no preamble." This one change saves 74% of output tokens. (Section V, T5)
2. **Do not rely on persona-based prompts for efficiency.** They improve quality, not cost. Role-based prompts increased output by 21.7% in my tests. (Section IX)
3. **Use Concise Chain-of-Thought for reasoning tasks.** Pair step-by-step reasoning with a conciseness instruction. This saves roughly 10% whilst keeping quality high. (Section V, T4; based on Wei et al., 2022)
4. **Keep it simple.** One well-chosen technique beats six stacked together. (Section VIII)
5. **Pick the right model.** Haiku handles most queries well at 20% of Opus's cost. Use Opus only where quality demands it. (Anthropic Documentation, 2026)

11.2 Implications for AI Agents

AI agents* make dozens or hundreds of API calls per task. Every call is an opportunity for format constraints to save tokens. If an agent makes 50 calls per task, T5 formatting saves roughly 74% on *each* call. For an enterprise running thousands of agent tasks per day, this compounds into significant savings in both cost and execution time.

11.3 Limitations

- **Sample size:** Five questions per configuration. A larger sample would strengthen confidence.
- **Quality not measured:** I measured token quantity, not response quality. Format-constrained answers are shorter but less detailed. Users should evaluate whether shorter answers meet their needs.
- **Single model family:** Results may differ for GPT-4, Gemini, or open-source models.
- **Single-turn only:** I tested one-off questions, not multi-turn conversations.

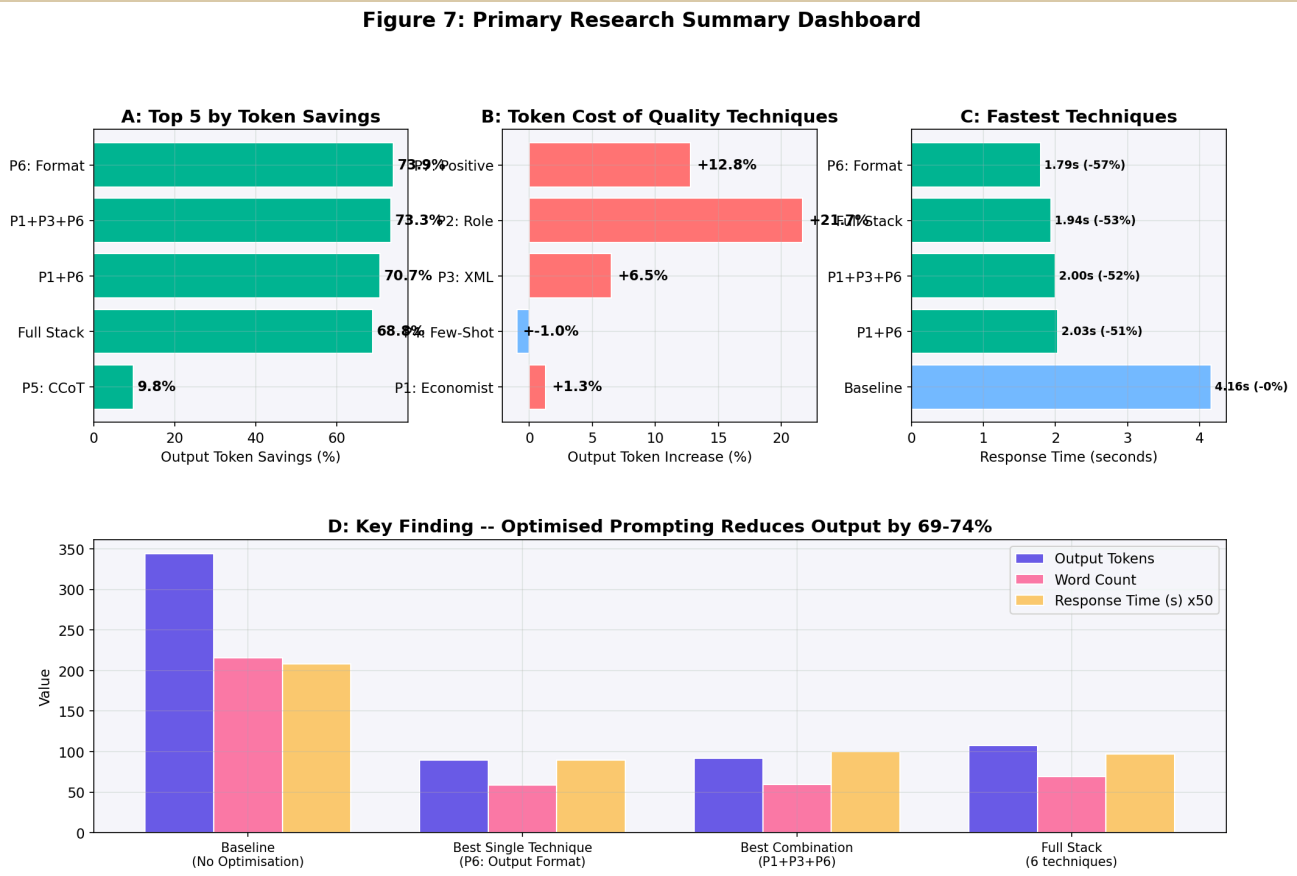

Figure 7. *Summary dashboard showing all key metrics.*

> **What this chart shows:** Four panels summarising the entire study. Panel A ranks techniques by token savings (Output Format wins at 73.9%). Panel B shows the "token cost" of quality techniques (Positive Framing adds 12.8% more tokens). Panel C ranks techniques by speed (Output Format is fastest at 1.79s). Panel D compares baseline, best single technique, best combination, and full stack side by side, making it clear that the single best technique outperforms all combinations.

"Every token you do not generate is a token you do not pay for, a watt you do not consume, and a drop of water you do not evaporate."

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CONNECT WITH THE AUTHOR

Samraj Matharu

Founder, The AI Lyceum

hello@theailyceum.com

For questions about this research, collaboration enquiries, or speaking requests.

Disclaimer

This paper presents the results of independent research conducted by The AI Lyceum. It is not sponsored by, endorsed by, or affiliated with Anthropic, OpenAI, Google, or any other AI company. All pricing data is sourced from publicly available documentation as of March 2026 (see Reference 1) and may change without notice. The cost projections are estimates based on experimental data and real-world results may vary depending on query complexity, response length, and model version.

The prompt techniques described here are provided for educational purposes. While my experiments measured token reduction, I did not assess response quality or factual accuracy. Users should evaluate whether format-constrained responses meet their specific needs before deploying these techniques in production.

This research was conducted using Claude Haiku 4.5. Results may differ on other models, including other Claude versions, GPT-4, Gemini, or open-source models.

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"The unexamined algorithm is not worth running."

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